

Serial Port WiFi Configuration Guide

1. Experiment Purpose

Connect to the control board over the serial port and use the configuration script to configure both the control board WiFi and the ESP32 camera module WiFi in one go.

2. Experiment Principle

1. Key Term Explanation

Keyword	Description
Serial port configuration	Communicates with the ESP32-S3 control board over the USB virtual serial port (UART) and sends binary configuration frames
Configuration frame protocol	A custom binary protocol: frame header (0xFF) + device ID (0xF8) + payload length + command code + payload + checksum
NVS	Non-Volatile Storage, the non-volatile storage of the ESP32 where parameters such as the WiFi password are persisted
config window	A configuration window of about 2 seconds after the firmware powers up; configuration commands are accepted only during this window
CAMERA_CONFIG_NOW	A dedicated protocol command relayed through the control board serial port that sends the WiFi information down to the ESP32 camera module

2. Technical Architecture

The configuration flow is relayed through the control board, so there is no need to connect the camera module directly:

```
PC (config_robot.py)
  | USB serial port (/dev/ttyUSB0 or /dev/microros_board)
  ▼
ESP32-S3 control board (firmware config window)
  | UART2
  ▼
ESP32 camera module (receives the WiFi config and connects)
```

Data flow: `config_robot.py` sends binary command frames over the serial port → the control board firmware parses them and writes to NVS / forwards them to the camera module → the camera module connects to WiFi and returns its IP → the script reads back the IP address with the `CAMERA_QUERY_IP_NOW` command.

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☒ Supported
No-PTZ version	☒ Supported
Required peripherals	ESP32 vision module, Type-C data cable

2. Prerequisites

1. The control board is connected to the virtual machine via a Type-C data cable
2. The camera module is installed on the car and connected to the control board with a ribbon cable
3. Confirm that the serial device exists: `ls -l /dev/microros_board` has output

3. Edit the Configuration Parameters

Editable parameters:

Parameter	Default	Description	Adjustment Suggestion
<code>main_wifi_ssid</code>	<code>SSID</code>	The Wi-Fi name the robot main controller connects to	Change it to an available 2.4G Wi-Fi; the ESP32-S3 does not support 5G Wi-Fi
<code>main_wifi_passwd</code>	<code>PASSWORD</code>	The Wi-Fi password the robot main controller connects to	Must correspond to the Wi-Fi name above
<code>agent_ip_addr</code>	<code>[192, 168, 33, 146]</code>	The IP address of the computer/virtual machine running the micro-ROS Agent	Change it to the real IP of the virtual machine or host on the same LAN; keep the format as 4 numbers
<code>agent_ip_port</code>	<code>8090</code>	The UDP port of the micro-ROS Agent	No need to change by default; if a different port was specified when starting the Agent, keep it consistent here
<code>ros_domain_id</code>	<code>77</code>	ROS_DOMAIN_ID	Must match the ROS_DOMAIN_ID in the virtual machine terminal
<code>ros_namespace</code>	<code>""</code>	ROS namespace	Keep it as an empty string for a single robot; fill it in only when multiple robots need to separate topics, e.g. <code>"robot1"</code>
<code>servo_1_offset</code>	<code>0</code>	S1 pan offset, in degrees	Fine-tune when the camera is not pointing straight ahead; positive corrects to the right, negative corrects to the left
<code>servo_2_offset</code>	<code>0</code>	S2 tilt offset, in degrees	Fine-tune when the camera pitch is off; positive corrects upward, negative corrects downward
<code>motor_pid</code>	<code>(1.8, 0.12, 0.0)</code>	Chassis motor speed loop PID	Normal users should not modify; change it only when motion control behaves abnormally and tuning is explicitly needed
<code>imu_yaw_pid</code>	<code>(2.4, 0.0, 0.16)</code>	Straight-line heading-hold PID	Normal users should not modify; change it only when the straight-line heading-hold performance needs optimization

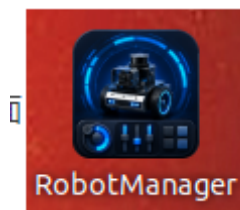
Servo offset adjustment notes:

- `servo_1_offset` adjusts the pan servo S1. When the camera points too far left, increase the value appropriately; when it points too far right, decrease the value appropriately.
- `servo_2_offset` adjusts the tilt servo S2. When the camera points too low, increase the value appropriately; when it points too high, decrease the value appropriately.
- It is recommended to make small adjustments of $1^{\circ} \sim 3^{\circ}$ each time, then re-run the configuration script and observe the effect.
- If the actual assembly direction is opposite to the behavior described above, adjust the offset values in the opposite direction based on the actual movement direction.

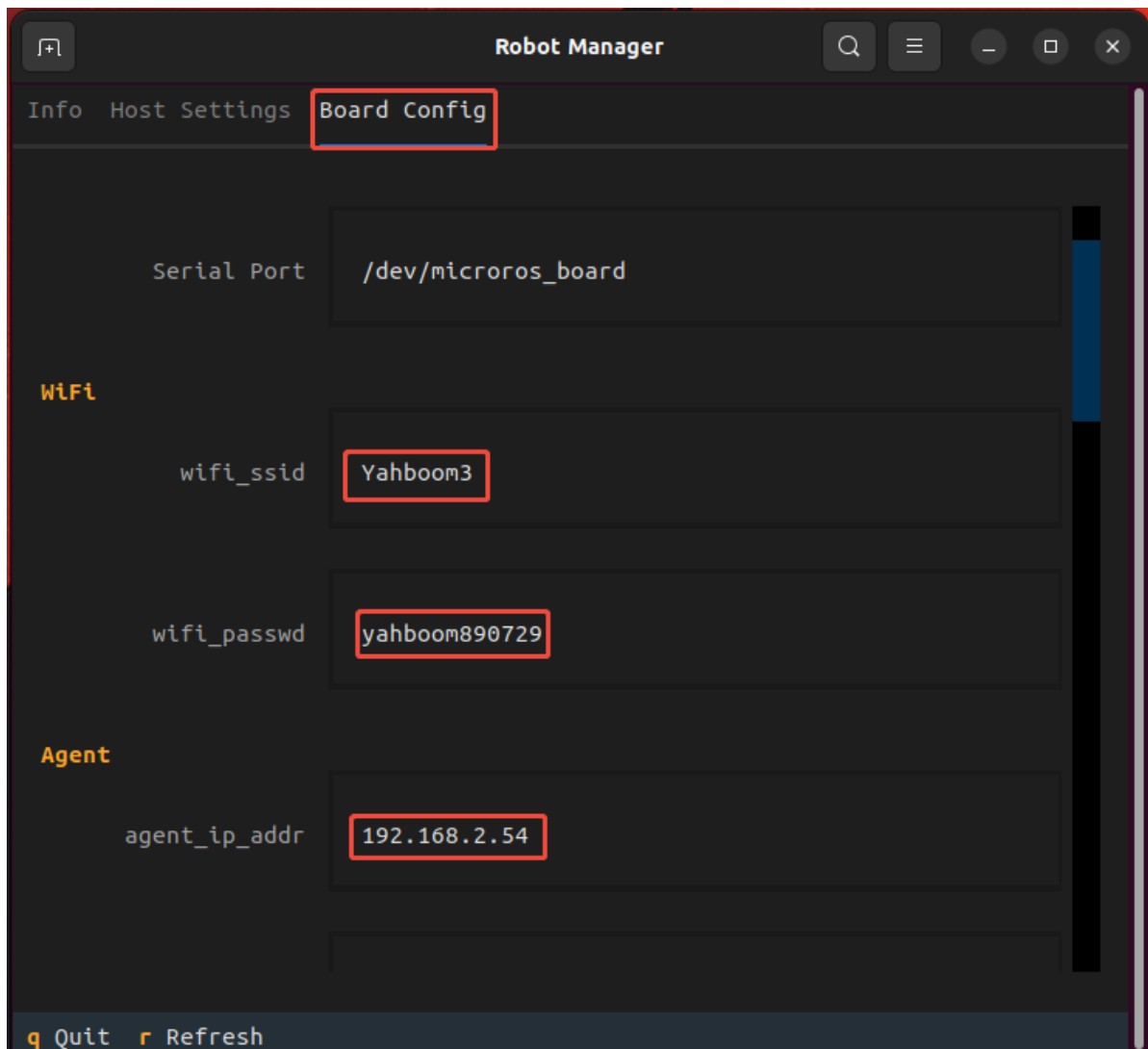
4. Run the Configuration

4.1 Configure the Control Board Parameters

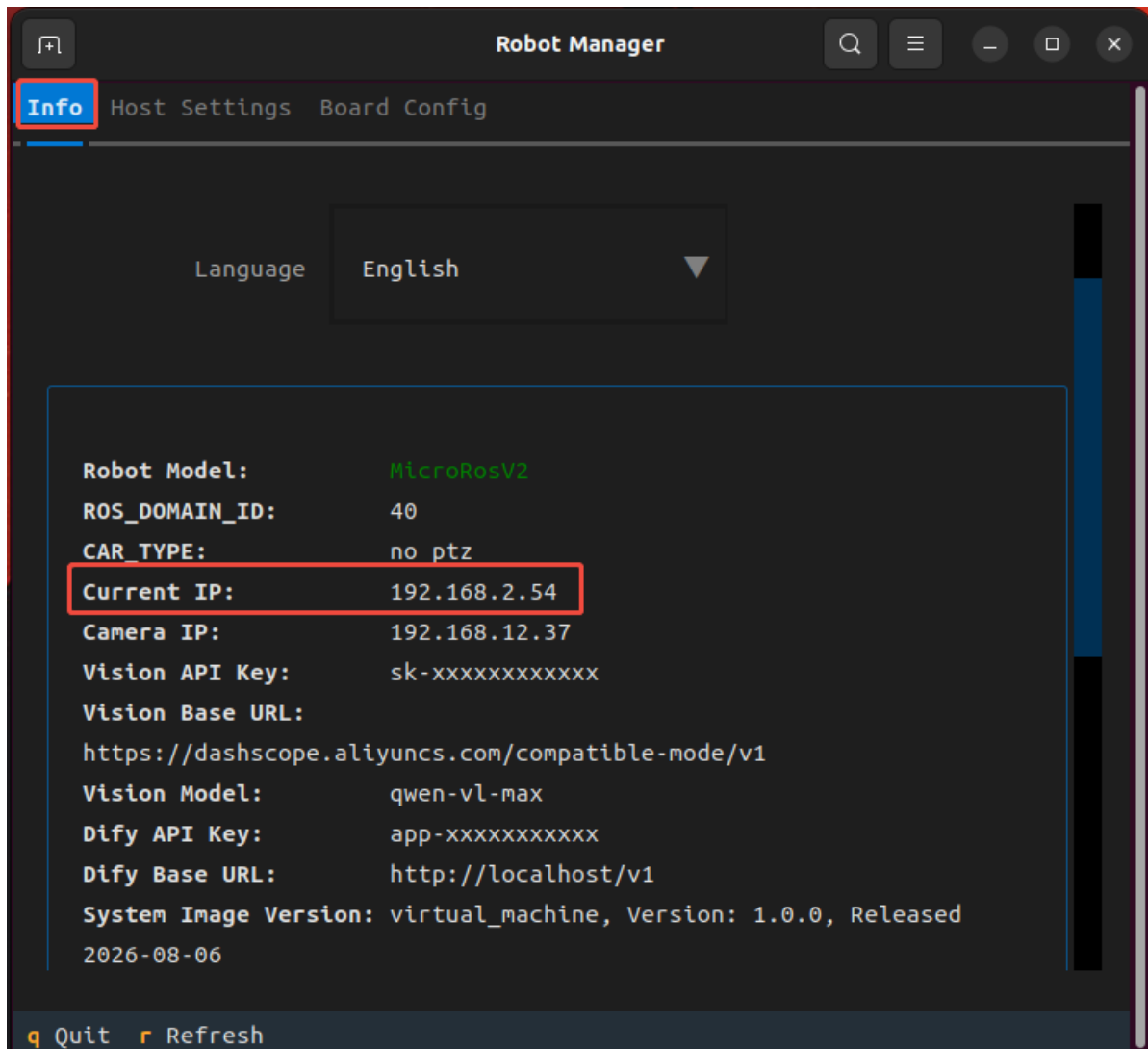
- Double-click the RobotManager application icon on the virtual machine desktop



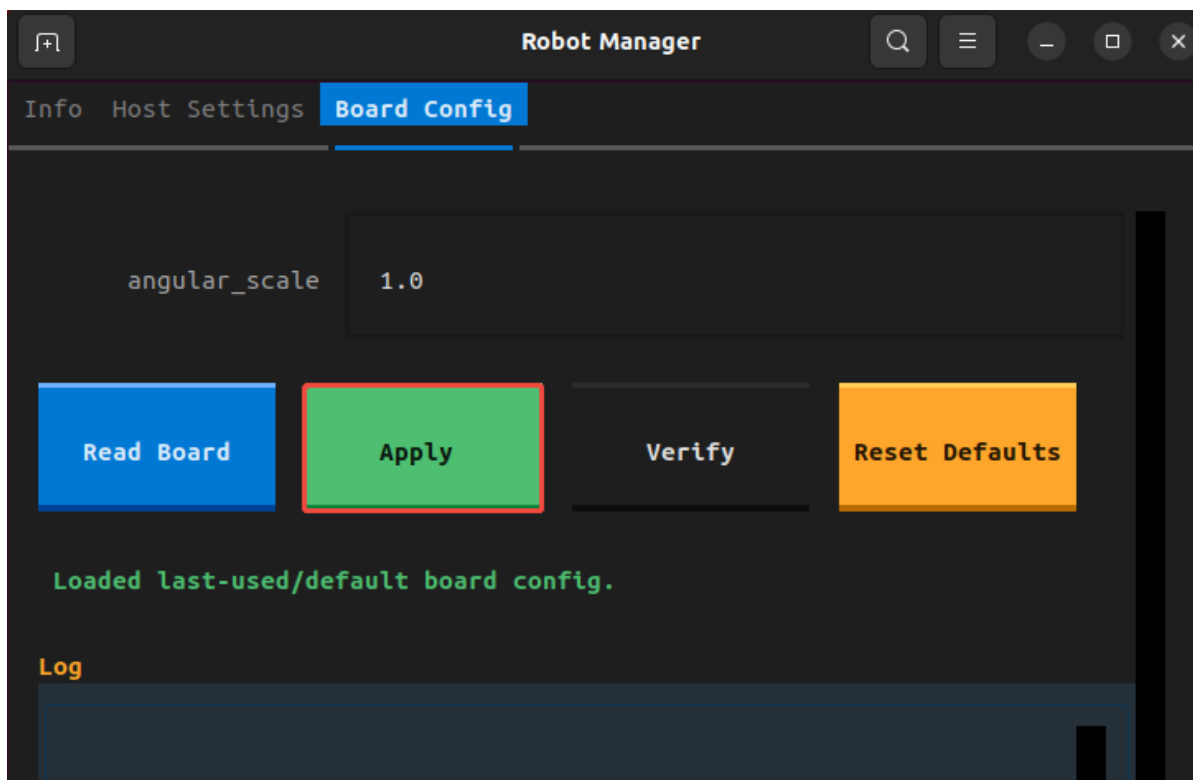
- Click the **Control Board** panel and fill in the WiFi name and password (make sure the board is on the same subnet as the virtual machine)



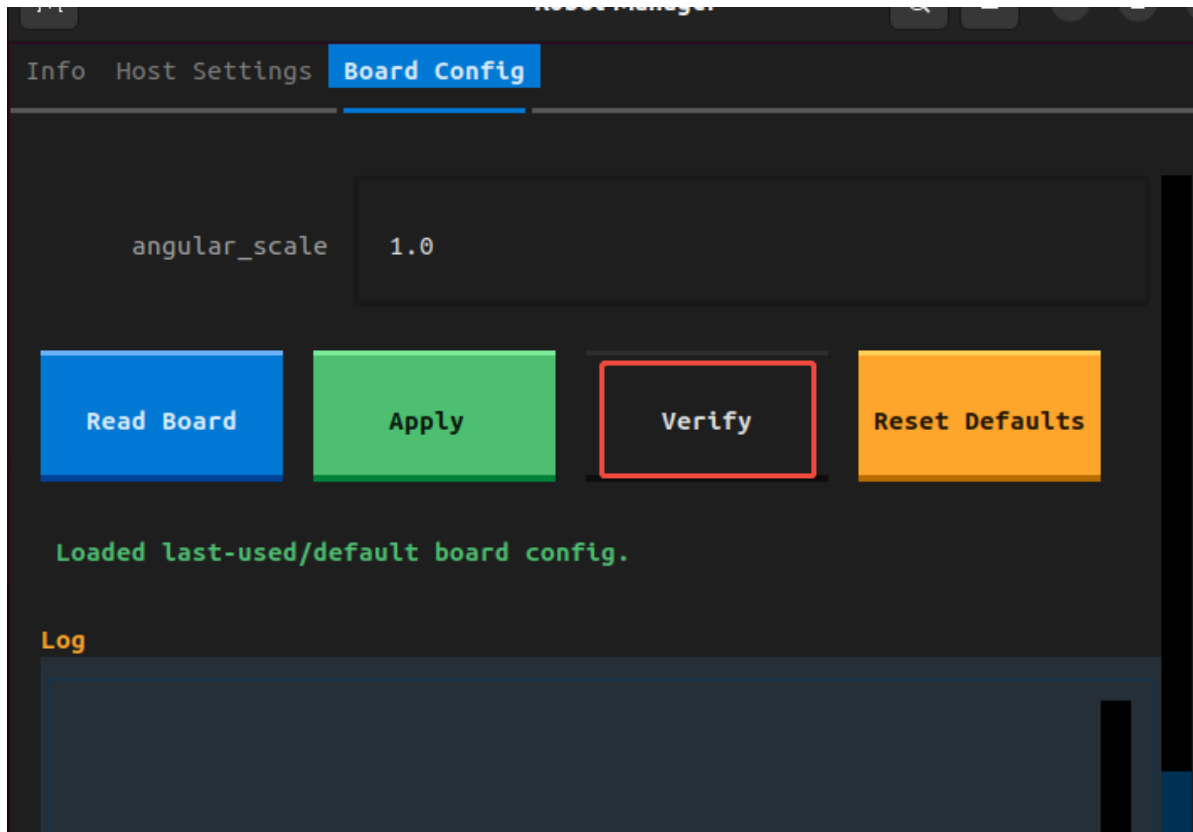
- Fill in `agent_ip_addr` with the virtual machine IP address; you can view the virtual machine IP address in the Info panel



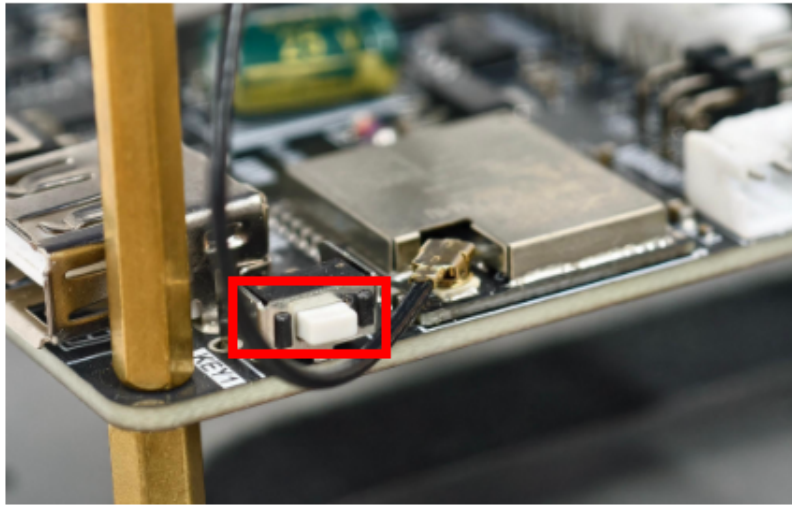
- Click Apply Configuration to write the above parameters to the MicroRos control board



- Click Verify to confirm the parameters match, and the control board parameter configuration is complete



- Press the button on the right side of the control board to switch the OLED screen display and view the information



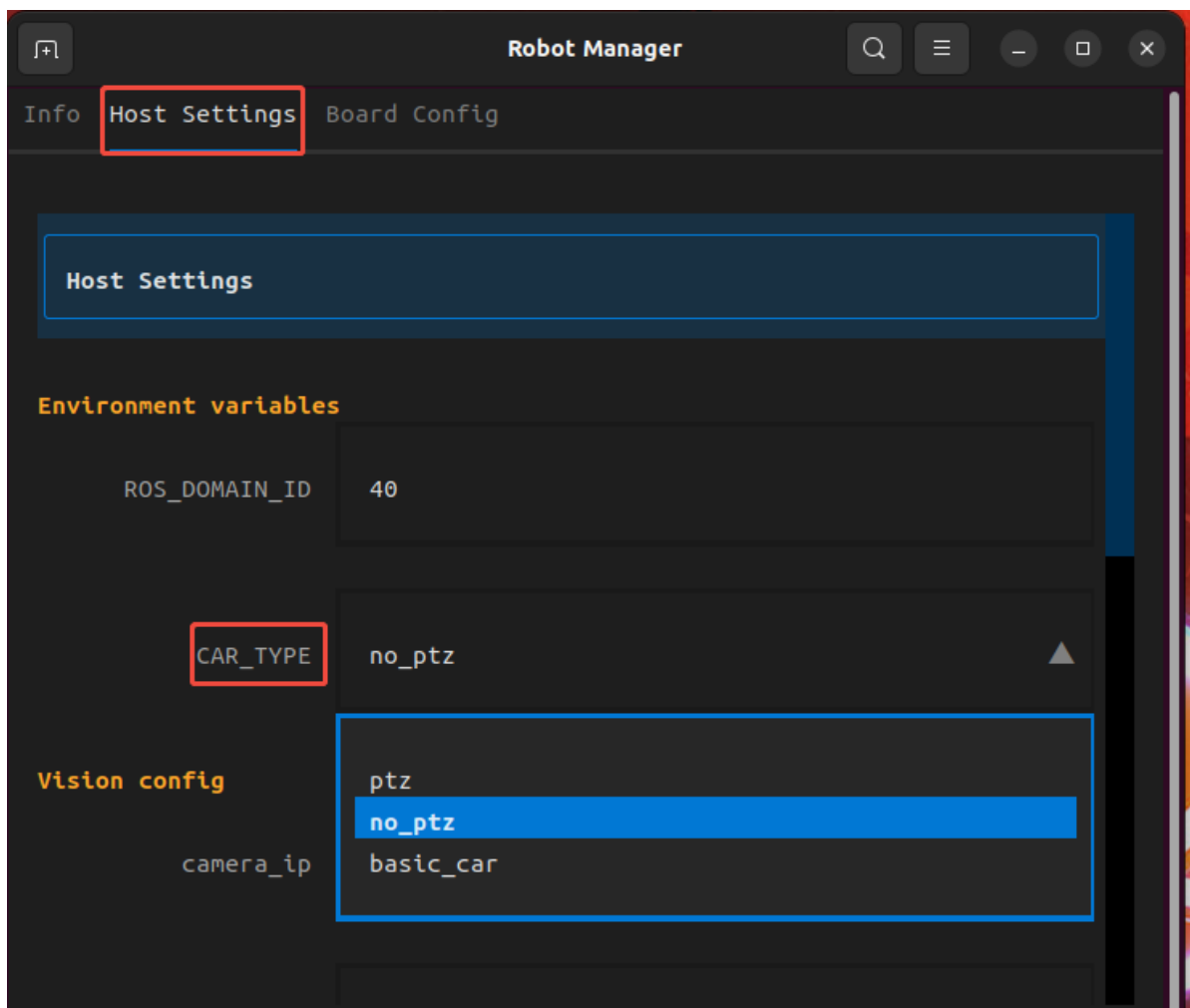
self: the vehicle's own IP address, make sure it is on the same subnet as the virtual machine

agent: the virtual machine IP address written in the previous step

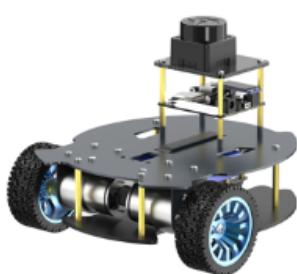
camera: the IP address of the image transmission camera

4.2 Configure the Vehicle Type and Camera IP in the Virtual Machine

- In the host settings, open the CAR_TYPE dropdown list and select the corresponding vehicle model



- The vehicle model names are as follows:



basic_car

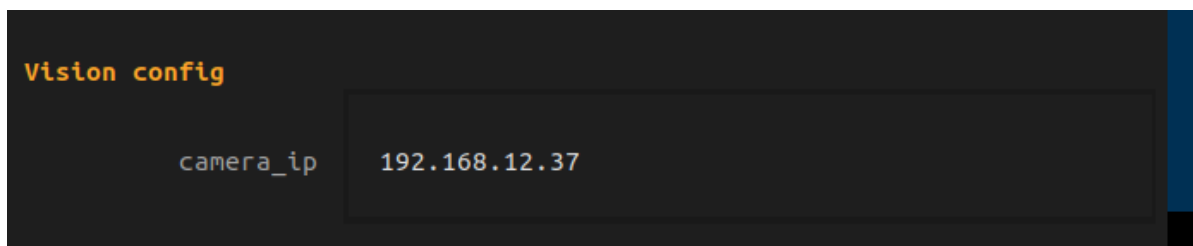


no_ptz



ptz

- For the camera IP configuration, fill in the camera IP address shown on the vehicle's OLED screen



- Click Save Configuration, then press **q** or simply close the window to exit **RobotManager**

Save

Reload

Saved. Env vars take effect in newly opened terminals.

q Quit r Refresh